

Level 1 ID Variation

Nominal Families

- Hold
- Circle
- Lissajous
- Waypoint (Test split)

Same Scenario
and Parameters

Level 2 Transferability

Cross-Evaluation

Evaluate native PID
from one family on another
(e.g., Circle PID
tested on Waypoint)

Classic Robustness

Level 3 Compositions

Composite Paths

Trajectories linked
sequentially
(Hold → Circle →
Waypoint)

Mixed OOD
Generalization

Level 4 OOD Geometries

New Geometries

- Lemniscates
- Complex Lissajous

Absolute OOD
Generalization

Increasing Degree of Novelty